Conditional Diffusion Models

Date

이전 내용) Diffusion model

$$x_T \xrightarrow{q_t / p_\theta} x_0, \qquad x_T \xleftarrow{q} x_0$$

목표) $L_{VLB} \downarrow \rightarrow p_\theta \approx q$

$$\nabla_\theta \left\| \epsilon_t(x_t|x_0) - \epsilon_\theta\left(\sqrt{\bar\alpha_t}\,x_0 + \sqrt{1-\bar\alpha_t}\,\epsilon, t\right) \right\|^2$$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar\alpha_t}}\,\epsilon_\theta(x_t,t)\right) + \sigma_t z$$

$$\epsilon_\theta(x_t,t) = \int \epsilon_t(x_t|x_0)\, q(x_0|x_t)\, dx_0$$

$$\nabla_{x_t} \log q(x_t|x_0) = \frac{-1}{\sqrt{1-\bar\alpha_t}}\,\epsilon_t(x_t|x_0)$$

$$\nabla_{x_t} \log q(x_t) = \frac{-1}{\sqrt{1-\bar\alpha_t}}\,\epsilon_\theta(x_t,t)$$

Classifier Guided Diffusion

∘ $\nabla_{x_t} \log p_\theta(x_t|y) = \frac{-1}{\sqrt{1-\bar\alpha_t}}\,\epsilon_\theta(x_t|y)$

$\rightarrow -\sqrt{1-\bar\alpha_t}\,\nabla_{x_t} \log p_\theta(x_t|y) = \epsilon_\theta(x_t|y) = \hat\epsilon(x_t)$

∘ $\nabla_{x_t} \log p(x_t|y) = \nabla_{x_t} \log \frac{p(y|x_t)\, p(x_t)}{p(y)}$

$= \nabla_{x_t} \log p(y|x_t) + \nabla_{x_t} \log p(x_t) + C$

$= \nabla_{x_t} \log p(y|x_t) + \frac{-1}{\sqrt{1-\bar\alpha_t}}\,\epsilon_\theta(x_t)$

$\times(-\sqrt{1-\bar\alpha_t})$

$-\sqrt{1-\bar\alpha_t}\,\nabla_{x_t} \log p(x_t|y) = \hat\epsilon(x_t) = \epsilon_\theta(x_t|y)$

$= -\sqrt{1-\bar\alpha_t}\,\boxed{\nabla_{x_t} \log p(y|x_t)} + \epsilon_\theta(x_t)$

Classifier

ex) $t=0, \dots$ 1000 이면

1000 단계의 classifier 필요.

DDIM에서)

$\hat\epsilon \leftarrow \epsilon_\theta(x_t) - \sqrt{1-\bar\alpha_t}\,\nabla_{x_t} \log p_\phi(y|x_t)$

$x_{t-1} \leftarrow \sqrt{\bar\alpha_{t-1}}\left(\frac{x_t - \sqrt{1-\bar\alpha_t}\,\hat\epsilon}{\sqrt{\bar\alpha_t}}\right) + \sqrt{1-\bar\alpha_{t-1}}\,\hat\epsilon$

ablated = ADM

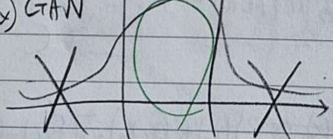
||

truncated trick

||

특정부분을 가중치, 나머지 부분은 자중.

ex) GAN



$$\bar\epsilon_\theta(x_t,t|y) = \epsilon_\theta(x_t,t) - \sqrt{1-\bar\alpha_t}\, w\, \nabla_{x_t} \log p(y|x_t)$$

w를 가변설정

1.5 ~ 3.5

w↑ → 성능↑ = fidelity ↑

w↓ → diversity ↑

Date

기존 DDPM) $p_\theta(x_t | x_{t+1}) = N(\mu, \Sigma)$

$\rightarrow \mu_\theta(x_t | x_{t+1}) = \frac{1}{\sqrt{\alpha_t}}\left(x_{t+1} - \frac{\beta_t}{\sqrt{1-\bar\alpha_t}}\epsilon_\theta(x_t, t)\right)$

Classifier DDPM) $p_\theta(x_t | x_{t+1}, y)$

$= N(\underline{\mu + w \cdot \Sigma \nabla_{x_t} \log p_\theta(y|x_t)}, \Sigma)$

pf) $p_\theta(x_t | x_{t+1}, y)$

$= \frac{p_\theta(y | x_t, \cancel{\boxed{x_{t+1}}})\, p_\theta(x_t | \boxed{x_{t+1}})}{p_\theta(y | \boxed{x_{t+1}})} \rightarrow Z$ (∵ 이 항은 x_t 과 무관)

x_{t+1}이 더 도움이 없으니 x_{t+1} 필요X

$= Z \cdot p_\theta(y | x_t) \cdot p_\theta(x_t | x_{t+1})$

log

$\log p_\theta(y|x_t) + \log p_\theta(x_t | x_{t+1})$ --- Ⓘ

$p_\theta(x_t | x_{t+1}) = N(\mu, \Sigma)$

$= \frac{1}{Z} \cdot e^{\frac{-1}{2}(x_t - \mu)^t \cdot \Sigma^{-1}(x_t - \mu)}$

log

$\log p_\theta(x_t | x_{t+1}) = \frac{-1}{2}(x_t - \mu)^t \cdot \Sigma^{-1}(x_t - \mu) + C$ Ⓘⓘ

테일러 정리) $f(x) \approx f(x_0) + (x - x_0) \cdot \nabla_x f(x_0)$

$x_0 \to \mu$

$\log p_\theta(y|x_t) \approx \log p_\theta(y|x_t) + (x_t - \mu) \cdot \nabla_x \log p_\theta(y|x_t)$ Ⓘⓘⓘ

$x_t \to \mu$, C_1, g

by Ⓘ, $\log p_\theta(y|x_t)$ (Ⓘⓘⓘ) $+ \log p_\theta(x_t | x_{t+1})$ (Ⓘⓘ)

$\approx C_1 + (x_t - \mu) \cdot g + \frac{-1}{2}(x_t - \mu)^t \Sigma^{-1}(x_t - \mu) + C$

$= \frac{-1}{2}(x_t - \mu)^t \Sigma^{-1}(x_t - \mu)$

$- \frac{2}{2}(-1)(x_t - \mu)^t \Sigma^{-1} \cdot \Sigma g + C_2$

$= \frac{-1}{2}(x_t - \mu - \Sigma g)^t \Sigma^{-1}(x_t - \mu - \Sigma g) + C_2$

$= \log p_\theta(x_t | x_{t+1}, y) + C_2$

$p_\theta(x_t | x_{t+1}, y) =$

$= C_0 \cdot e^{\frac{-1}{2}(x_t - (\mu + \Sigma g))^t \cdot \Sigma^{-1} \cdot (x_t - (\mu + \Sigma g))}$

$= N(\mu + \Sigma g, \Sigma)$

$\mu_\theta(x_t | x_{t+1}, y)$

$= \underbrace{\mu_\theta(x_t | x_{t+1})}_{\mu} + \underbrace{\Sigma}_{\Sigma} \cdot \underbrace{\nabla_{x_t} \log p_\theta(y|x_t)}_{g}$

← ADM

$= \mu + \boxed{w} \Sigma \cdot g$

$= \mu_\theta(x_t | x_{t+1}) + \boxed{w} \Sigma \cdot \nabla_{x_t} \log p_\theta(y|x_t)$ $\boxed{\text{III}}$

classifier base

$p_\theta(x_t|x_{t+1}, y)$ → $N(\mu + \Sigma \nabla \log p(y|x_t), \Sigma)$ → x_{t-1} · x_t ~

$\epsilon_\theta(x_t, t|y)$

classifier free

x_t → UNet → ϵ_θ → $\hat{\epsilon}_\theta$ → x_0

$\epsilon_\theta(x_t, t|y) \rightarrow \hat{\epsilon}_\theta(x_t, t|y)$

GLIDE에서) 방금 만든 $\mu_\theta(x_t|x_{t+1}, y)$

강화

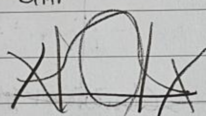
$$\hat{\mu}_\theta(x_t|y) = \mu_\theta(x_t|y) + s \cdot \Sigma_\theta(x_t|y) \nabla_{x_t} \log p_\theta(y|x_t)$$

원래 이건데 강화

W↓ → diversity ↑ 다양한 그림 fidelity ↓ 의미있는부분, 성능 퀄리티

W↑ → diversity ↓ fidelity ↑

≈ Truncation Trick in GAN



◦ Inception Score = IS

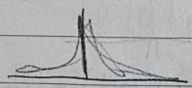
IS = fidelity + Diversity

IS↑ → Good↑

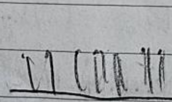
$$\ln IS = H\left[E\left[p_{cls}(\cdot|x)\right]\right] + E\left[H\left[p_{cls}(\cdot|x)\right]\right]$$

Diversity fidelity

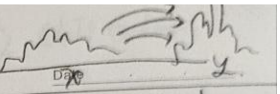
ex) fidelity ↑ → H↓ diversity ↓



fidelity ↓ diversity ↑ → H ↑



$IS = e^{\square}$



◦ FID = 차원과의 거리

차원그림과 생성그림의 차이 ↓ → 성능↑

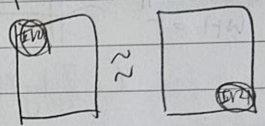
유표) $\|x-y\|^2$ ↓

ex) Wasserstein 거리

$$d_F(\mu, \nu) = \left(\inf \int \|x-y\|^2 \, d\gamma(x, y)\right)^{\frac{1}{2}}$$

◦ Perceptual loss

문제)



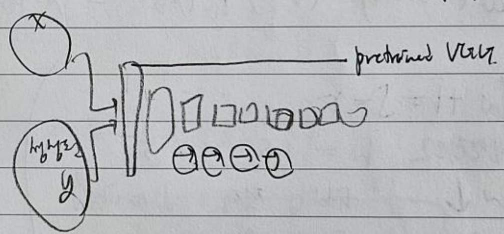
둘 원래 같은 img인데

pixel 단위로 하면 다르다고 판단되는게 문제

→ VGG 안에서의 ⊖ ⊖ 이게 서로 같으면 두 사진이 같다고 판단하는것

= Perceptual loss.

ex) SRGAN = super Resolution GAN] pretrained V



Classifier free diffusion

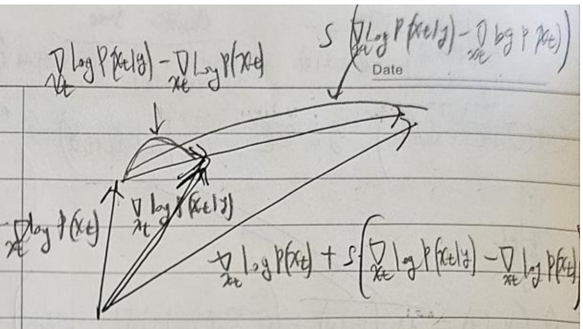
구조) ① $\epsilon_\theta(x_t|y) = \epsilon_\theta(x_t, t) - w\sqrt{1-\bar{\alpha}_t} \nabla_{x_t} \log p_\theta$

x_t → U-NET → $\epsilon_\theta(x_t, t|y)$

노이즈

조절 → $\hat{\epsilon}_\theta(x_t, t|y)$ → x_{t-1}

- - - → x_0



증폭된

$$\hat{\epsilon}_\theta(x_t, t|y) = \epsilon_\theta(x_t, t) + s\cdot\left(\epsilon_\theta(x_t|y) - \epsilon_\theta(x_t, t)\right)$$

방금 Unet으로 얻는것

$$\boxed{\hat{\epsilon}_\theta(x_t|y)} = \epsilon_\theta(x_t, t) + \boxed{s}\left(\epsilon_\theta(x_t|y) - \epsilon_\theta(x_t, t)\right)$$

$w+1 = r$

$$C_1\cdot\nabla_{x_t}\log p(x_t|y) - C_2\nabla_{x_t}\log p(x_t) + \nabla_{x_t}\log p(y)$$

$$= \nabla_{x_t}\frac{p(x_t|y)\cdot p(y)}{p(x_t)}$$

$$= \nabla_{x_t}\log p(y|x_t)$$

Classifier가 됨

$\tilde{\epsilon}_\theta(x_t, t, y) = (w+1)\cdot\epsilon_\theta(x_t, t, y) - w\cdot\epsilon_\theta(x_t, t)$ 으로도 표현가능.

∘ $w+1 = s = r$

일반적으로 $w = 1.5 \sim 3.5$

$w\downarrow$) fidelity 적당. diversity↑

w 중간) fidelity ↑

$w\uparrow$) fidelity↑ diversity↓

$$\epsilon_\theta(x_t|y) = \nabla_{x_t}\log p(x_t|y) = \nabla\log p(x_t) + r\cdot\underbrace{\nabla\log p(y|x_t)}_{\nabla\log p(x_t|y) - \nabla\log p(x_t)}$$

다른 표현

$$= \nabla\log p(x_t) + r\nabla\log p(x_t|y) - r\nabla\log p(x_t)$$

$$= r\nabla\log p(x_t|y) + (1-r)\nabla\log p(x_t)$$

$w = r-1$

$$\tilde{\epsilon}_\theta(x_t, t, y) = (w+1)\,\epsilon_\theta(x_t, t, y) - w\,\epsilon_\theta(x_t, t)$$

최신

Unet으로 얻은 구형 ϵ_θ

준비   성능) Classifier free > CLIP Guidance.

Date

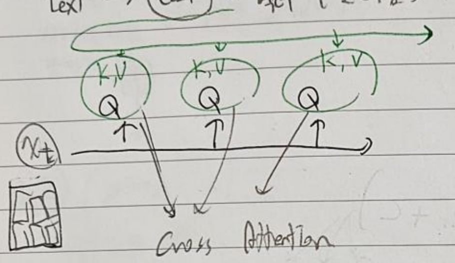
- Unet 구조 Cross Attention

$x_t \rightarrow$ U-Net $\rightarrow \epsilon_\theta(x_t, t|y)$

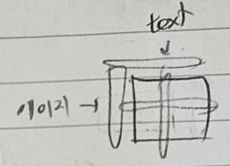
$\hookrightarrow \hat{\epsilon}_\theta(x_t, t|y) \rightarrow x_{t-1}$

$\cdots \rightarrow x_0$

text → CLIP → 임베딩 벡터 (관련이미지 고려됨)



예) CLIP



같은 이미지, text 쌍이면 내적한게 1이 되도록 학습시키는 인코더.
출력은 임베딩 벡터.

K.V : text를 tokenization 해서 씀.

Q : x_t를 ViT처럼 tokenization 해서 Q로 사용.

- Masked 아님.
- Positioned index는 그냥 지정.

GLIDE : CLIP Guidance

GLIDE) $\hat{\mu}_\theta(x_t|y) = \boxed{\mu_\theta(x_t|y)} + s \cdot \Sigma_\theta(x_t|y) \boxed{\nabla_{x_t} \log p_\theta(y|x_t)}$ 강화.

DDPM) $\boxed{\mu_\theta(x_t|x_{t+1}, y)} = \mu_\theta(x_t|x_{t+1}) + w \cdot \Sigma \nabla_{x_t} \log p_\theta(y|x_t)$

CLIP Guidance) $\hat{\mu}_\theta(x_t|c) = \mu_\theta(x_t|c) + s \cdot \Sigma_\theta(x_t|c) \boxed{\nabla_{x_t}(f(x_t) \cdot g(c))}$

$\nabla_{x_t} \log$ 함

$\nabla_{x_t} \log p_\theta(c, x_t) = \nabla_{x_t} \log p_\theta(x_t|c) + \cancel{\nabla_{x_t} \log p(c)}$

$\parallel$

$\nabla_{x_t} \log \frac{e^{f(x_t) \cdot g(c)}}{A} = \boxed{\nabla_{x_t} f(x_t) \cdot g(c)}$

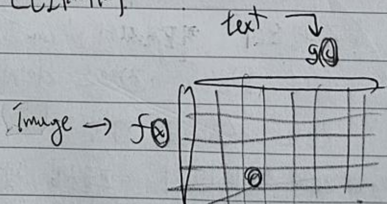
$\boxed{\nabla_{x_t} \log p_\theta(y|x_t)} = \nabla_{x_t} \log p_\theta(c|x_t) = \underbrace{\nabla_{x_t} \log p_\theta(c, x_t)}_{} \underbrace{\nabla_{x_t} \log p(x_t)}_{}$

CLIP Guidance 가지 보려서

∨ 둘이는 제거

CLIP Guidance er

예) CLIP에서

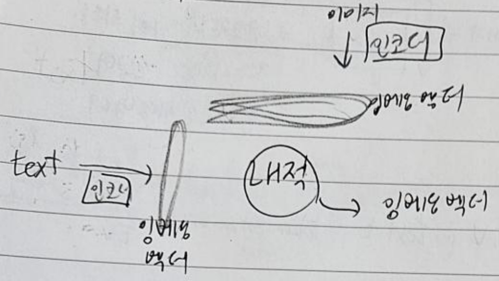


$p_\theta(c_i|x_i) = \frac{e^{f(x_i) \cdot g(c_i)}}{\sum_{i=1}^{N} e^{f(x_i) \cdot g(c_i)}}$

DALLE 2 : UnCLIP 클립 뒤집기

Date

· 클립 CLIP



· 안 3쿤의 DALLE 2 설명

↳ 텍스트는 순서정보 기반 정보, 문맥적 의미가 중심.

↳ 이미지는 위치적 연속성 포함, 시각적 특징 중심

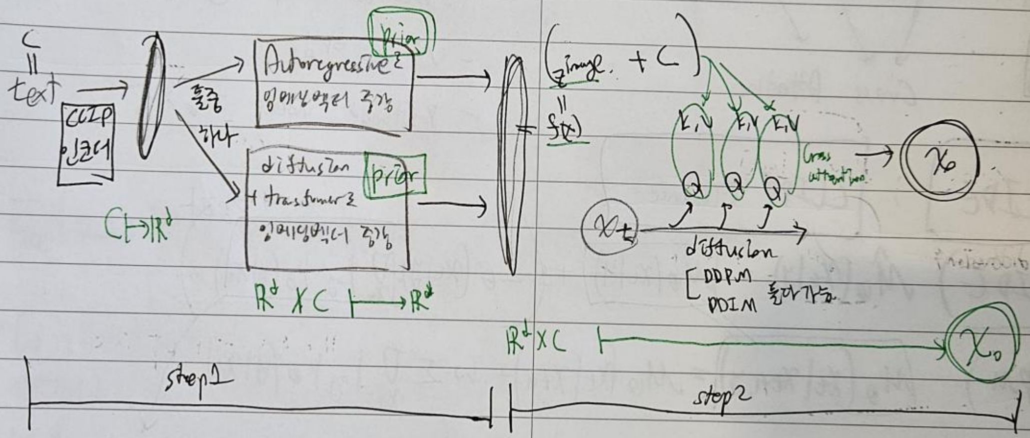
↳ 구조적 차이 ⊙ → 완벽히 정렬되기 어렵다.

↳ 잠재공간의 분포차이 ⊙

↳ CLIP은 단순히 text와 이미지를 비교.

but Prior를 추가해 더 정교한 정렬 가능.

· DALLE 2 특징 / 구조



step1) Case 1) Autoregressive model

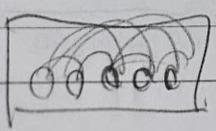
· 느림,

세부적 차이 중요시 여길때 굿.

· 정밀도가 필요할 때.

· 텍스트 임베딩을 조건으로 이미지 임베딩 요소를 순차적으로 예측.

· 대규모 데이터에서 효율성 낮아질 수 있음.



Case 2) Diffusion Prior

· 더 빠른 생성 속도, 다채로운 결과물이 요구될 때

· 이미지 임베딩에 점진적으로 추가된 노이즈를 제거하면서 생성.

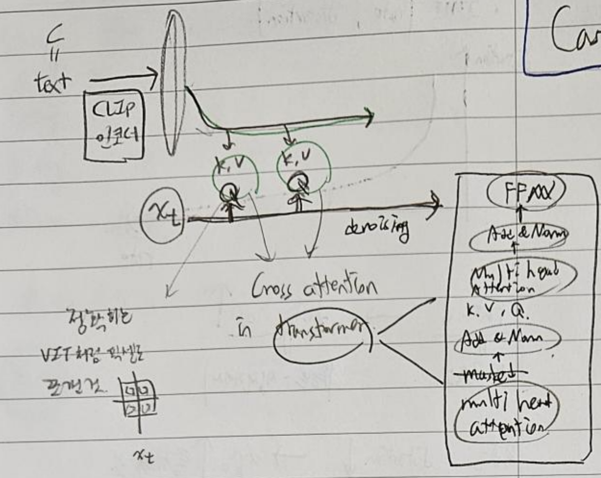
· 샘플링 속도가 상대적으로 빠름.

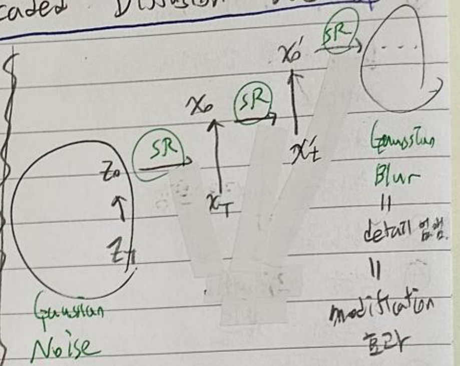
· 분포를 탐색하며 다양한 출력 생성 가능.

· Autoregressive 보다는 세밀하지 X.

Date

Cascaded Diffusion Model





Gaussian Noise
||
다양성 + 흔들음,
흔얼개
||
overfitting 방지.
||

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$
||
$$N(x) = N(x,y)$$

step 2) (z^{image}, c)

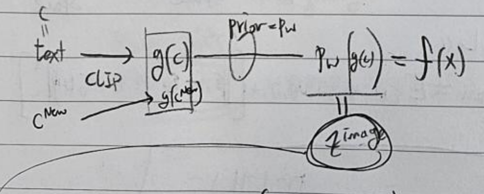
Case 1) c만 있다

Case 2) c X z^{image} O

Case 3) c, z^{image} 둘다 O

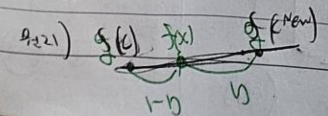
diffusion은 Cascaded (super resolution) 쓰기도 함.

+) Image Manipulations (이미지 합성)



$$z^{image} = f(x) + h\left(g(c^{new}) - g(c)\right)$$
$\quad \|$
$p_w(g(c))$

사후분포 ——— 이게 DALLE-2 핵심
↓
성능↑

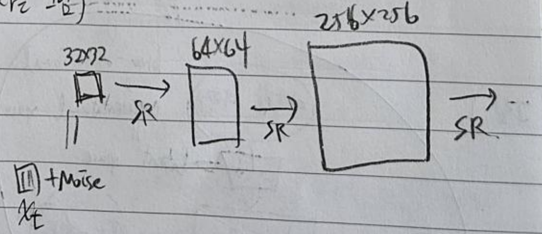


d) 수식

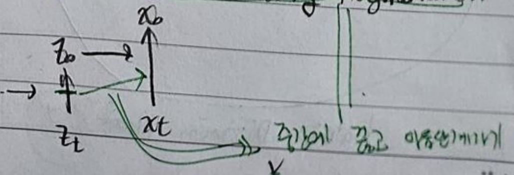
$$E_{(x,y)} E_{\epsilon, r} \left\| f_\theta\left(\boxed{x}, \sqrt{r}\, y_0 + \sqrt{1-r}\,\epsilon, r\right) - \epsilon \right\|^p$$

(f_θ = ϵ_θ, $\sqrt{r}\, y_0 + \sqrt{1-r}\,\epsilon$ = y_t, r = t)

다른 그림)



d) Truncated Conditioning Augmentation



· 특징

① 저해상도 이미지 생성

↓ 점진적으로 업스케일링

고해상도로 변환

② 노이즈 첨가

: 추가적인 세부정보 보강

③ 생성 품질 향상

④ 학습을 독립적으로,

모듈화 가능.

⑤ DALLE-2

Imagen

Stable Diffusion

) 예시.

Stable Diffusion

→ pixel space가 아닌,

latent space 가 있다는게 특징

↑

이미지 압축시킨거 = 의미부분

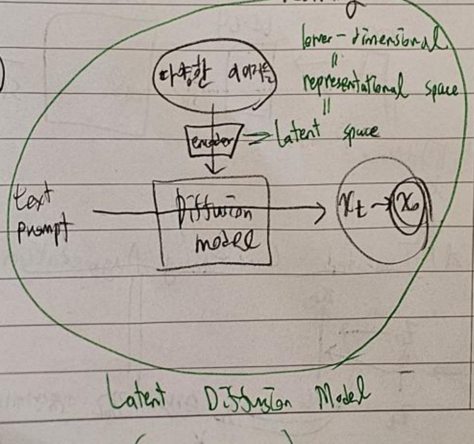
$\parallel$

perceptual details compression

$\parallel$

semantic compression = generative modeling learning

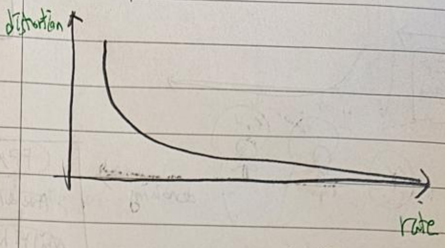
그림)



Latent Diffusion Model

(LDM)

· 그래프 (rate, distortion)



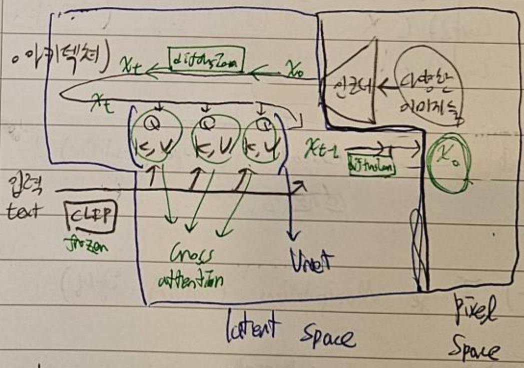
$\underbrace{\text{rate}\downarrow}_{\text{차원축소}} \rightarrow \underbrace{\text{distortion}\uparrow}_{|\text{원본} - \text{생성이미지}|}$

목표는 distortion ↓ → 성능 ↑ 좋게하는건.

즉, 차원축소는 문제가 없으니

그걸 걍 encoder로 latent space에 보내버리자.

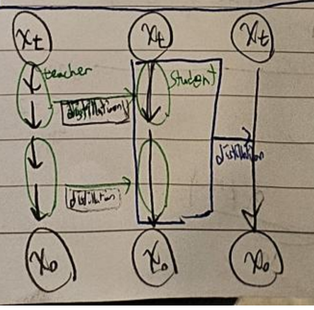
· 아키텍처)



· Loss

$$L_{LDM} = E_{\varepsilon(x), y, \epsilon \sim N(0,1), t}\left[\| \epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y)) \|_2^2\right]$$

Progressive Distillation



ㄷ) ⓓ : denoising = DDIM

알고리즘에서)

많은그림 = $X_0(x_t)$: ⓓ가 N번

distillation을 : $\tilde{x}(z_t)$

마칠때

: ⓓ가 $\frac{N}{2}$번